Entering the Age of Agents

A Strategic Roadmap for Making Neuroelectrics Technologies Agent-Ready

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RE: Agent-Ready Architecture — Cloud APIs, MCP Integration, Neurotwin-as-a-Service

1. Executive Summary

The healthcare AI market is undergoing a structural shift from co-pilot tools to autonomous AI agents that can observe, reason, plan, and act across clinical workflows. The Model Context Protocol (MCP), introduced by Anthropic in late 2024 and now governed by the Linux Foundation, is rapidly becoming the standard interface through which these agents connect to external tools, data sources, and devices. Major LLM providers — OpenAI, Google DeepMind, Microsoft, AWS — have all adopted it. Healthcare MCP adoption stands at approximately 32% and rising.

Neuroelectrics is exceptionally well-positioned for this transition. Our Neurotwin digital twin framework, StimWeaver optimization engine, and combined EEG/tES hardware platform represent a uniquely deep computational stack in neuromodulation. However, our current software integration layer — NIC2, MatNIC2, and the C-based SDK — was designed for the desktop/lab era. It is synchronous, locally-bound, and assumes a human operator at every step. This architecture cannot serve the agent-native world that is arriving now.

> ***Core thesis:*** *Neuroelectrics must build a cloud-native API layer and an MCP-compatible agent interface to ensure that our devices and computational models are accessible to the AI agents that will increasingly orchestrate clinical research, treatment delivery, and patient monitoring workflows. The companies that expose their capabilities through agent-ready interfaces will capture the platform position; those that don't will become commodity hardware suppliers.*

■ What is an AI agent?

An AI agent is software powered by a large language model (like ChatGPT or Claude) that can **autonomously plan and execute multi-step tasks**. Unlike a chatbot that only answers questions, an agent can read a patient's EEG history, reason about treatment options, call external tools, and present recommendations — all without a human manually orchestrating each step. Think of it as a very capable intern that can use all your software tools, read all your data, and propose actions for your approval.

■ What is an agentic API?

A traditional API lets one piece of software talk to another — for example, a weather app calling a weather service. An **agentic API** is designed specifically so that AI agents can discover what tools are available, understand what each tool does, and call them autonomously. The key difference: a human developer doesn't need to write custom code for every integration. The agent reads the API's description and figures out how to use it on its own. Making NE's devices 'agent-ready' means building APIs that agents can find and use without custom engineering.

2. Current State: The Integration Gap

Our current software stack was built for a world where a researcher sits at a computer running NIC2, manually configures a protocol, monitors an EEG session in real time, and exports data files for offline analysis. This model served us well for a decade. It will not serve us for the next five years. Here is a clear-eyed assessment of the gap:

Component	Current State	Agent-Era Requirement
Device Control	NIC2 desktop GUI, MatNIC2 (Matlab)	RESTful Cloud API with WebSocket streaming; MCP tool definitions
Data Access	Local file export (EDF+, ASCII, NEDF); LSL/TCP streaming	Cloud data lake with FHIR-compatible resource endpoints; real-time SSE/WebSocket feeds
Protocol Design	NIC2 UI manual configuration	Programmatic protocol definition via API; agent-callable StimWeaver optimization
Neurotwin	Research pipeline (internal)	Neurotwin-as-a-Service: API for twin generation, simulation queries, treatment prediction
Multi-Site Mgmt	Per-site NIC2 instances; manual coordination	Centralized cloud fleet management; programmatic device provisioning and monitoring
SDK	C libraries, 32-bit Windows/Linux	Python-first SDK; TypeScript SDK; cloud-native, cross-platform
Authentication	Device-level pairing	OAuth2 with scoped permissions (read:eeg, write:protocol, execute:stim); API keys; audit logs

The practical consequence is already visible. Our research customers build their own ad-hoc integration layers via LSL and TCP. CROs managing multi-site trials (consider the scale of ADASTRA: 232 patients across 5 clinical sites plus IfADo) are forced to coordinate via spreadsheets and email rather than programmatic

orchestration. This is friction we are imposing on our own users.

3. Strategic Roadmap: Three Horizons

Horizon 1: Cloud API Layer (6–12 Months)

"Make the Data Accessible"

The foundational prerequisite. We must expose the core capabilities currently locked inside NIC2 and the SDK as a set of documented, versioned, cloud-native REST APIs. This is not a rewrite of NIC2 — it is a new layer that sits above our existing device firmware and communicates with devices via the existing Bluetooth/WiFi transport, mediated through a lightweight gateway service.

> **■ What is a cloud API?**
>
> An API (Application Programming Interface) is a set of rules that lets software programs talk to each other. A **cloud API** means that instead of needing NIC2 running on a local laptop, any authorized software anywhere in the world can request data from NE's servers over the internet — for example: 'give me the last 10 EEG sessions for patient X' or 'start a stimulation protocol on device Y.' A **REST API** is the most common flavor, using standard web protocols (the same technology behind every website). A **WebSocket** is a persistent connection that streams data in real time, like a live EEG feed.

Enobio Cloud API

- EEG session management: create, start, stop, query sessions
- Historical data retrieval with filtering (time range, channels, frequency bands)
- Real-time EEG streaming via WebSocket or Server-Sent Events
- Device telemetry: battery, impedance, connection status, firmware version
- Marker and event management (insert, query, export)
- Data export in multiple formats: JSON, EDF+, FHIR-wrapped resources

Starstim Cloud API

- Protocol definition endpoints: montage design, waveform parameterization
- Session execution and real-time monitoring
- Stimulation dosage tracking and cumulative exposure queries
- Sham/active profile management for blinded trials
- **StimWeaver-as-API:** expose the montage optimization engine as a callable endpoint

Biomarker Intelligence Layer

Raw EEG data is necessary but not sufficient for agent consumption. Most AI agents cannot reason meaningfully over 32-channel microvolt time series. What they need are semantically rich, clinically validated derived features — biomarkers — served through the same API. This is where Neuroelectrics builds proprietary value on top of the hardware.

The Biomarker API exposes three tiers of derived intelligence from EEG recordings:

Tier 1 — Classical spectral features (real-time capable):

- Band power (delta, theta, alpha, beta, gamma) per channel and region

- Alpha/theta ratio, beta asymmetry (depression-relevant lateralization index)
- Power spectral density and peak alpha frequency
- Coherence and phase-locking value between channel pairs
- Signal quality and artifact probability scores

Tier 2 — Clinical composite biomarkers:

- EEG complexity measures (Lempel-Ziv, permutation entropy) — relevant to consciousness level and AD progression
- Seizure probability score (epilepsy pipeline) — continuous risk index from NE's proprietary detection algorithms
- Network connectivity metrics (functional connectivity matrices, graph-theoretic measures)
- Event-related potential (ERP) component extraction (P300, N400, MMN)
- Stimulation response index — quantified EEG change attributable to tES intervention, enabling treatment tracking

■ What is a foundation model?

A **foundation model** is a large AI model pre-trained on a massive dataset so that it learns general patterns, which can then be applied to specific tasks. GPT-4 is a foundation model for language. In our context, an **EEG foundation model** would be pre-trained on Starlab's large EEG corpus so that it learns the general structure of brain electrical activity. Clients could then run their own patient's EEG through this model to get meaningful features — without needing to train their own AI from scratch.

■ What is a latent space?

When a foundation model processes an EEG recording, it compresses the complex raw signal into a compact set of numbers called a **latent representation** (or embedding). The 'space' formed by all possible representations is the **latent space**. The magic: similar brain states end up near each other in this space. So a clinician could ask 'show me patients whose brain activity looks most similar to this patient' and get meaningful answers — not by comparing raw EEG squiggles, but by measuring distance in latent space. Think of it as a GPS coordinate system for brain states.

Tier 3 — Latent space representations:

This tier draws on a triad of complementary capabilities. BCOM (Barcelona Computational Foundation), a non-profit scientific partner founded by NE's CSO, provides the theoretical frameworks, computational model architectures, and AI guidance that define *what* to compute and *why*. Starlab Barcelona, NE's sister company specializing in EEG services, contributes the large curated EEG corpus on which foundation models are trained. Neuroelectrics productizes the result as a platform service:

- Clients choose from available EEG foundation models (trained on Starlab's large-scale corpus, architectures designed under BCOM scientific guidance) to generate latent space coordinates for their data
- Patient state vectors in a continuous latent space — enabling similarity search across patient populations
- Trajectory embeddings tracking patient state evolution over treatment courses

• Anomaly detection in latent space — flagging sessions that deviate from expected treatment response patterns

• Model versioning and selection API: clients specify which foundation model to apply, enabling research reproducibility and comparison across model architectures

> ***The NE – BCOM – Starlab triad:*** *Three entities, three distinct roles, one scientific vision.* ***BCOM*** *is the Scientist — a non-profit foundation providing computational model architectures, theoretical guidance, and AI support for the neuroscience sector.* ***Starlab*** *is the EEG Specialist — a sister company to NE that sells EEG services and has assembled one of the field's largest curated EEG corpora.* ***Neuroelectrics*** *is the Clinical Device Company — it builds the cloud platform that productizes BCOM's science and Starlab's data into agent-ready, clinically validated biomarker services. The strategic value lies in offering clients a platform where they select a foundation model, run their own EEG through it, and receive clinically meaningful latent coordinates — all without their raw data leaving their control.*

Real-time biomarker streaming (Tier 1 and select Tier 2 features) via WebSocket enables a critical capability: AI agents can monitor brain state during stimulation and trigger adaptive protocol adjustments. This is the agent-native evolution of the closed-loop paradigm that MatNIC2 Pro supports today in Matlab.

NE Portal Cloud (Management Plane)

• API key provisioning and rotation

• Device fleet registration and lifecycle management

• OAuth2 scoped access: `read:eeg`, `write:protocol`, `execute:stimulation`

• Audit logging and compliance reporting (GDPR, MDR, 21 CFR Part 11)

• Data residency controls (EU-hosted as default, with options)

> ***Business case:*** *A cloud API unlocks three immediate revenue opportunities: (1) API-access licensing for research institutions and CROs; (2) programmatic multi-site trial management for pharma partners; (3) platform integration fees from digital health companies embedding NE data in their clinical workflows.*

Horizon 2: MCP Server & Agent Interface (12–18 Months)

"Make the Devices Agent-Callable"

With the cloud API in place, the next step is to wrap it in an MCP Server that makes Enobio and Starstim directly accessible to AI agents through the Model Context Protocol. MCP is rapidly becoming healthcare AI's standard integration layer, analogous to what REST was for web services a decade ago. Any AI agent — regardless of which LLM powers it — can discover and call MCP-exposed tools.

> **■ What is MCP (Model Context Protocol)?**

MCP is an open standard — think of it as **USB-C for AI**. Just as USB-C lets you plug any device into any computer without a special cable, MCP lets any AI agent connect to any data source or tool through a single standard interface. MCP was created by Anthropic in 2024 and is now governed by the Linux Foundation, with adoption by OpenAI, Google, Microsoft, and AWS.

How MCP actually works: the protocol mechanics

Understanding the mechanics is essential for the Product team. MCP is not magic — it is a structured JSON protocol that sits between natural language and our internal API. Here is exactly what happens when a clinician asks an AI agent a question about a patient's EEG:

Step 1 — The clinician speaks natural language:

> *"Show me the alpha power from patient K-2847's session this morning."*

Step 2 — The LLM translates natural language into a structured JSON-RPC call:

The AI agent (Claude, ChatGPT, etc.) reads the clinician's request and figures out which MCP tool to call and what parameters to use. It generates a JSON-RPC message — **not** natural language — and sends it to the NE MCP Server:

```
{
  "jsonrpc": "2.0",
  "method": "tools/call",
  "params": {
    "name": "get_biomarkers",
    "arguments": {
      "session_id": "sess-K2847-20260407",
      "biomarker_type": "spectral"
    }
  }
}
```

Step 3 — The NE MCP Server translates JSON-RPC into an internal API call:

Our MCP server receives the JSON, parses it, and translates it into a call to our internal Cloud API — whatever form that takes (REST, database query, microservice):

```
GET https://api.neuroelectrics.com/v1/biomarkers
  ?session_id=sess-K2847-20260407
  &type=spectral
```

Step 4 — Data flows back through the same chain:

Our Cloud API returns the biomarker data as JSON. The MCP Server wraps it in a JSON-RPC response and sends it back to the agent. The LLM reads the structured data and presents it to the clinician in plain language: *"Patient K-2847's alpha power this morning was 45.3 μV², which is 12% below their 4-week baseline. Theta/alpha ratio is elevated at 1.41."*

The key insight: *The MCP Server is a thin translation layer — a public JSON interface that any agent can call. It does not contain business logic; it translates between the standardized JSON-RPC protocol (which all LLMs speak) and our proprietary internal API (which only we control). We define the tools and parameters once in the MCP Server; every AI agent in the world can then use them without custom integration. The tool definitions — what parameters are accepted, what data comes back — are our public specification. The internal API behind it is private.*

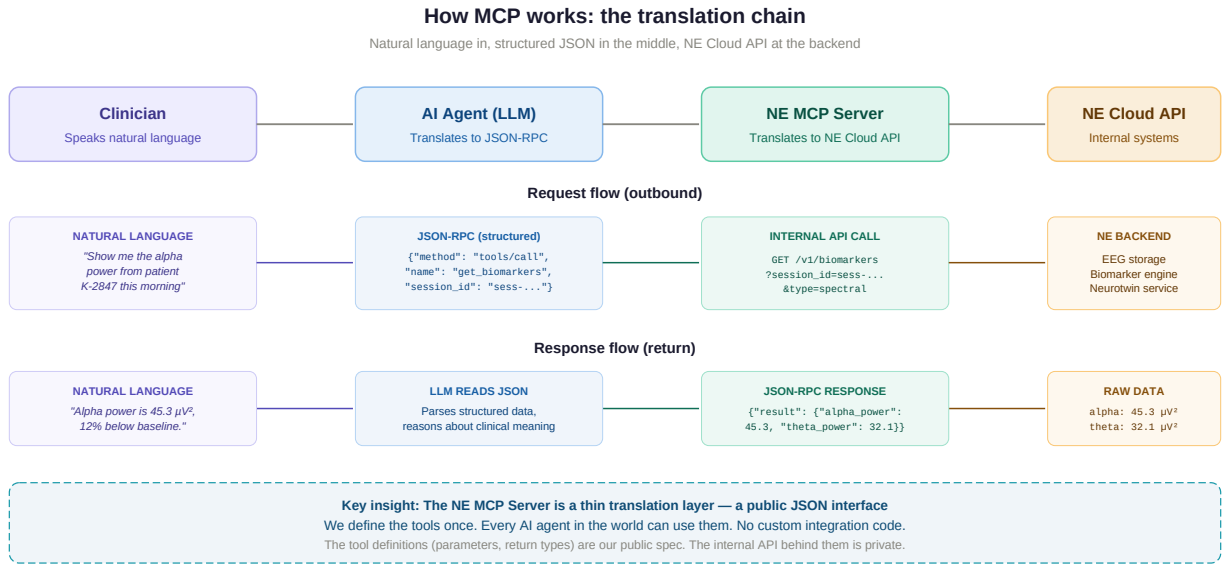


Figure 1: The MCP translation chain. Natural language enters from the left, is converted to structured JSON-RPC by the LLM, translated to internal API calls by the NE MCP Server, and data flows back through the same chain. The clinician never sees the JSON.

■ What is OAuth2 and scoped permissions?

OAuth2 is the standard way to grant limited access to a system without sharing passwords — it is the technology behind 'Sign in with Google.' When we say **scoped permissions**, we mean that each API key or agent gets access only to specific actions: for example, a research agent might get `read:eeg` (can view EEG data) but not `execute:stimulation` (cannot trigger treatment). This is how we maintain clinical safety in an agent-accessible system: the agent can suggest, but only authorized clinicians can approve actions that affect patients.

Proposed MCP Tool Definitions

MCP Tool	Function	Safety Class
`get_eeg_session`	Retrieve structured EEG data for a patient/session	Read-only
`get_neurotwin_model`	Return a patient's digital twin state and parameters	Read-only

MCP Tool	Function	Safety Class
`optimize_montage`	Call StimWeaver with target map and constraints; return ranked montage options	Compute (no device action)
`simulate_protocol`	Run a candidate stimulation protocol against the Neurotwin; return predicted outcomes	Compute (no device action)
`get_treatment_history`	Return full dosage and outcome trajectory for a patient	Read-only
`schedule_stimulation`	Queue a treatment session for clinician approval	Write (requires clinician confirmation)
`get_device_status`	Return device fleet health: battery, impedance, firmware	Read-only
`get_biomarkers`	Retrieve computed biomarker features (band power, coherence, complexity, seizure score) for a session or time range	Read-only
`stream_biomarkers`	Real-time WebSocket stream of Tier 1/2 biomarker features during active EEG recording or stimulation	Read-only (real-time)
`get_latent_embedding`	Return patient's learned latent state vector; supports similarity queries across patient cohorts	Read-only (premium)

The safety classification column is critical. MCP's architecture naturally supports tiered permissions: read-only tools are freely callable by agents, compute tools require authentication, and write/execute tools require explicit clinician-in-the-loop confirmation. This maps directly onto the regulatory requirement under MDR and FDA guidance that therapeutic device actions must involve physician oversight.

Agent Workflow Example: End-to-End Translation Chain

Consider a clinical AI agent managing a depression treatment program. The following shows the complete chain for one step — from clinician's natural language through JSON-RPC to NE's internal API and back:

Clinician says:

> *"How has this patient's brain activity changed over the last month of treatment?"*

The LLM reasons: "I need the patient's recent EEG sessions, then their biomarkers, then I'll compare trends." It breaks this into a sequence of MCP tool calls:

Call 1 — LLM sends JSON-RPC to NE MCP Server:

```
{"method": "tools/call", "params": {"name": "get_biomarkers",
  "arguments": {"session_id": "sess-latest", "biomarker_type": "spectral"}}}
```

→ MCP Server translates to: `GET /v1/biomarkers?session_id=sess-latest&type=spectral`
→ NE Cloud API returns spectral data → MCP Server wraps in JSON-RPC response → LLM receives structured data

Call 2 — LLM requests the baseline comparison:

```
{"method": "tools/call", "params": {"name": "get_biomarkers",
  "arguments": {"session_id": "sess-4-weeks-ago", "biomarker_type": "spectral"}}}
```

Call 3 — LLM requests the latent embedding trajectory:

```
{"method": "tools/call", "params": {"name": "get_latent_embedding",
  "arguments": {"session_id": "sess-latest", "model": "ne-eeg-foundation-v1"}}}
```

LLM synthesizes and responds in natural language:

> *"Over the past month, alpha power at F3 has increased 18% (from 38.2 to 45.1 μV²), and the interhemispheric asymmetry index has normalized from 0.31 to 0.12. The patient's latent embedding has shifted 0.34 units toward the healthy-volunteer cluster. These trends are consistent with a positive treatment response. I recommend continuing the current tDCS protocol for another 4 weeks."*

Notice what happened: the clinician asked **one question in plain English**. The agent autonomously made **three structured JSON-RPC calls** to NE's MCP Server, each of which was translated into an **internal NE Cloud API request**. The clinician never saw the JSON. The agent handled the orchestration. NE's MCP Server handled the translation. NE's Cloud API handled the data.

Horizon 3: Neurotwin-as-a-Service (18–36 Months)

"The Brain Digital Twin as an AI Backend"

The ultimate strategic position: Neuroelectrics becomes the computational neurostimulation backend that any AI healthcare platform can plug into. The analogy is Stripe for payments — you don't build your own stimulation physics engine, you call NE's API.

The Neurotwin program (Horizon 2020 EIC Pathfinder, coordinated by NE Barcelona) has produced personalized hybrid brain models that combine electromagnetic physics with neuronal dynamics. The tES 2.0 framework introduces simulations of how neurons — not just tissues — respond to applied electric fields. This is a unique computational capability in the market. The strategic imperative is to productize it.

NTaaS (Neurotwin-as-a-Service) Pipeline

- **Ingest:** Patient MRI + EEG baseline uploaded via API
- **Segment:** Automated MRI segmentation (7-tissue model: skin, skull, CSF, GM, WM, eyes, ventricles) using NE's deep neural network pipeline
- **Build:** Generate personalized biophysical head model with tissue-specific conductivities
- **Calibrate:** Fit neural mass model parameters to patient's EEG dynamics
- **Simulate:** Expose twin for protocol simulation queries (predict network response to candidate stimulation)
- **Optimize:** Run StimWeaver against twin-informed targets
- **Refine:** Post-treatment EEG fed back to update twin parameters (the write-back loop)

Strategic positioning: *NTaaS transforms Neuroelectrics from a medical device company into a computational neuroscience platform company. Device sales become the hardware layer of a recurring-revenue software and computation business. Every patient treated with NE hardware generates a Neurotwin that becomes more valuable over time — a flywheel of data, models, and clinical evidence.*

4. Market Demand: What Customers Will Want

By Customer Segment

Customer Segment	Current Need (2026)	Emerging Need (2027–2029)
CROs / Pharma	Multi-site device provisioning; centralized EEG data aggregation; automated compliance monitoring; API-based trial coordination	AI agent-managed trials; autonomous protocol adaptation; real-time safety signal detection via agent workflows; Neurotwin-powered virtual patient cohorts for trial design
Health Systems / Clinics	EHR-integrated neuromodulation data (FHIR); clinician-facing dashboards; treatment outcome tracking	Clinical AI agents with read access to NE device data; ambient documentation of stimulation sessions; agent-recommended treatment adjustments
Digital Health Platforms	Wearable-grade API integration; MCP plug-in for brain health data alongside other physiological streams	Brain-health AI coaching; EEG-informed sleep/cognitive optimization; consumer neuromodulation guided by AI
Research Labs	Python-first SDK replacing MatNIC2; cloud data storage; versioned, documented APIs	AI-assisted experiment design; agent-driven closed-loop paradigms; computational reproducibility via API-logged sessions

5. Regulatory Considerations

Moving to cloud-connected, agent-accessible medical devices introduces new regulatory dimensions. These are manageable but must be addressed from the architecture phase, not retrofitted.

■ What is FHIR?

FHIR (Fast Healthcare Interoperability Resources) is the global standard for exchanging healthcare data electronically. It defines how patient records, lab results, observations, and device data should be structured so that different hospital systems, apps, and devices can understand each other. If NE's EEG data is FHIR-compatible, a hospital's electronic health record system can display it alongside blood tests and imaging results — without custom software. Think of it as a common language for healthcare IT.

■ What is SaMD (Software as a Medical Device)?

SaMD is a regulatory classification for software that performs a medical function on its own — without being part of a physical device. If our cloud API simply stores and transmits EEG data, it may not be SaMD. But if StimWeaver-as-API analyzes patient data and recommends a treatment protocol, that software *is* the medical device, and it needs regulatory clearance (CE marking in EU, FDA clearance in US) just like Starstim itself. The key question is: does the software influence a clinical decision?

EU MDR + AI Act

Under the EU AI Act (implementation by 2026), AI systems intended for medical use are classified as high-risk. Any AI agent that interacts with NE devices for therapeutic purposes will fall under this classification. NE's MCP interface design must include quality management documentation, transparency requirements, and human oversight mechanisms as built-in architectural features — not add-ons.

FDA / SaMD Considerations

The cloud API layer and Neurotwin service will likely constitute Software as a Medical Device (SaMD) under FDA's digital health framework. StimWeaver-as-API in particular — if it provides treatment recommendations based on patient data — crosses the SaMD threshold. Early engagement with FDA's Digital Health Center of Excellence is advisable. The predetermined change control plan (PCCP) pathway may be relevant for the Neurotwin's continuously-learning model architecture.

Cybersecurity

Cloud-connected therapeutic devices face heightened cybersecurity scrutiny. Architecture must include zero-trust principles, encrypted device-to-cloud communication, API rate limiting, and anomaly detection. Healthcare remains the most expensive sector for data breaches. Our API security posture must meet HITRUST and/or NIST frameworks from day one.

6. Technical Requirements Summary

Requirement	Specification	Priority
Cloud API framework	OpenAPI 3.1 spec; REST + WebSocket; versioned endpoints	P0 (Critical)
Authentication	OAuth2 + PKCE; API keys; scoped permissions; MFA for write operations	P0
Data formats	JSON primary; EDF+ export; FHIR R4 resource wrappers for EEG observations	P0
MCP Server	MCP-compliant tool + resource definitions; JSON-RPC transport; stdio + SSE	P1 (High)
Device gateway	Edge service bridging BLE/WiFi device transport to cloud API; local-first with cloud sync	P1
Python SDK	Pythonic wrapper around Cloud API; pip-installable; async support; typed	P1

Requirement	Specification	Priority
Hosting / Infra	EU-hosted (GCP europe-west1 or AWS eu-west-1); Kubernetes; SOC 2 Type II target	P1
FHIR Integration	FHIR R4 Observation resources for EEG; DiagnosticReport for session summaries	P2 (Medium)
NTaaS Pipeline	Containerized MRI segmentation + twin generation; GPU compute for simulation	P2
Biomarker Engine	Real-time spectral/coherence pipeline; trained latent-space models; GPU inference for Tier 3; versioned model registry	P1
Audit / Compliance	Immutable audit logs; 21 CFR Part 11 electronic signatures; GDPR data subject rights API	P1

7. Recommended Immediate Actions

The following actions can be initiated in Q2 2026 with existing resources, providing foundations and proof-of-concept demonstrations for the full roadmap.

1. Software stack audit. Map every function in NIC2 and MatNIC2 as either business logic (API-extractable) or presentation logic (UI-only). This produces the definitive scope for the Cloud API.

2. OpenAPI specification draft. Write the Enobio and Starstim Cloud API specification before writing any code. An OpenAPI spec is a shareable, reviewable artifact that can be socialized with pharma partners, CROs, and strategic investors immediately.

3. MCP proof-of-concept. Build a minimal MCP server (read-only) that wraps existing NE data access. Demonstrate an AI agent (Claude or GPT) querying a patient's EEG history in natural language. This demo has outsized impact for investor and grant narratives.

4. FHIR working group engagement. Engage with the HL7 FHIR community on EEG and neuromodulation resource definitions. FHIR lacks mature neuromodulation resources — NE can help define the standard and gain first-mover credibility.

5. Neurotwin narrative reframe. Update external communications to position the Neurotwin not as a concluded EU research project but as the foundation of a production-grade AI backend for personalized brain stimulation. The science is published; the product wrapper is the gap.

6. Regulatory pre-assessment. Commission a gap analysis for the cloud API and NTaaS under MDR SaMD classification and the EU AI Act high-risk requirements. Early clarity here prevents costly rework.

8. Competitive Context

No direct competitor in the non-invasive neuromodulation space has yet built a cloud-native, agent-accessible API layer. This window is open but will not remain so indefinitely. Consumer EEG companies (Muse/Interaxon, Emotiv, Neurable) are already exposing cloud APIs and SDK integrations, though none operate in the therapeutic/medical-grade space that NE occupies. The risk is not that a neuromodulation competitor builds this first — it is that a large digital health platform (e.g., one backed by a major tech or pharma company) builds a proprietary neuromodulation stack that bypasses NE entirely because NE's hardware was not API-accessible.

> ***The strategic imperative is clear:*** *build the integration layer before someone else builds around us. The company that owns the computational interface to personalized brain stimulation — not just the electrodes — will own the category.*

9. Quick Win: Staking the Claim

The full roadmap spans 36 months. But the marketing opportunity is immediate. No competitor in the non-invasive neuromodulation space has claimed the agent-ready positioning. The first credible move wins the narrative. Here is a concrete plan executable within 6–8 weeks:

The Deliverable: NE MCP Server (Demo Edition)

Build a lightweight, open-source MCP server that exposes read-only access to sample EEG session data from Enobio recordings. This does not require the full cloud API — it wraps existing exported data files (EDF+, CSV) in MCP tool definitions that any AI agent can discover and call.

What it includes:

- **3–4 MCP tools:** `list_sessions`, `get_eeg_data`, `get_biomarkers`, `get_device_info` — wrapping a curated sample dataset of anonymized EEG recordings
- **Published on GitHub** under NE's org, with MIT or Apache 2.0 license
- **pip-installable** Python package (`pip install neuroelectrics-mcp`)
- **Works immediately** with Claude Desktop, ChatGPT, Cursor, and any MCP-compatible client
- **Demo dataset** of 5–10 anonymized Enobio 32 sessions with pre-computed Tier 1 biomarkers

The Demo Video (90 seconds)

Screen recording showing a researcher opening Claude Desktop, connecting the NE MCP server, and having a natural-language conversation with their EEG data: "Show me the alpha power trend across sessions for this patient." "Compare the coherence patterns before and after stimulation." "Which session showed the strongest treatment response?" The agent calls NE's MCP tools, retrieves structured data, and presents insights conversationally. This is viscerally compelling and trivially reproducible by any viewer with an Enobio.

The Landing Page

A new page on neuroelectrics.com: **"Agent-Ready Brain Technology."** Positions NE as the first medical-grade neuromodulation company to offer MCP-native device integration. Links to the GitHub repo, the demo video, and a waitlist for the full Cloud API (Horizon 1). The waitlist captures lead intent from CROs, pharma, digital health platforms, and research labs — precisely the customer intelligence needed to prioritize Horizon 1 features.

Marketing Amplification

• **Blog post:** "Why We Built the First MCP Server for Brain Stimulation" — technical credibility piece for the research and developer community

• **LinkedIn / X announcement** from NE leadership and BCOM channels, targeting the AI-in-healthcare and digital health audiences

• **Developer community seeding:** submit to MCP server directories, Awesome MCP lists, and HuggingFace; engage with the Anthropic and OpenAI developer communities

• **Conference demo:** live MCP agent demo at next relevant neuromodulation or digital health event

Why this works as a quick win: *The MCP demo server costs minimal engineering effort (weeks, not months), uses existing data, requires no regulatory approval (it is a developer tool wrapping anonymized sample data, not a medical device), and generates outsized marketing value. It positions NE at the intersection of neuroscience and AI agents — a narrative no competitor currently occupies. The waitlist for the full API converts attention into actionable sales pipeline.*

Appendix: System Architecture Diagram

The following diagram illustrates the target architecture for the Future NE Cloud System, showing the relationships between client interfaces (clinicians, devices, patients), the core cloud services (EEG storage, patient data repository, Neurotwin engine, biomarker intelligence layer), and the external agent access points via MCP.

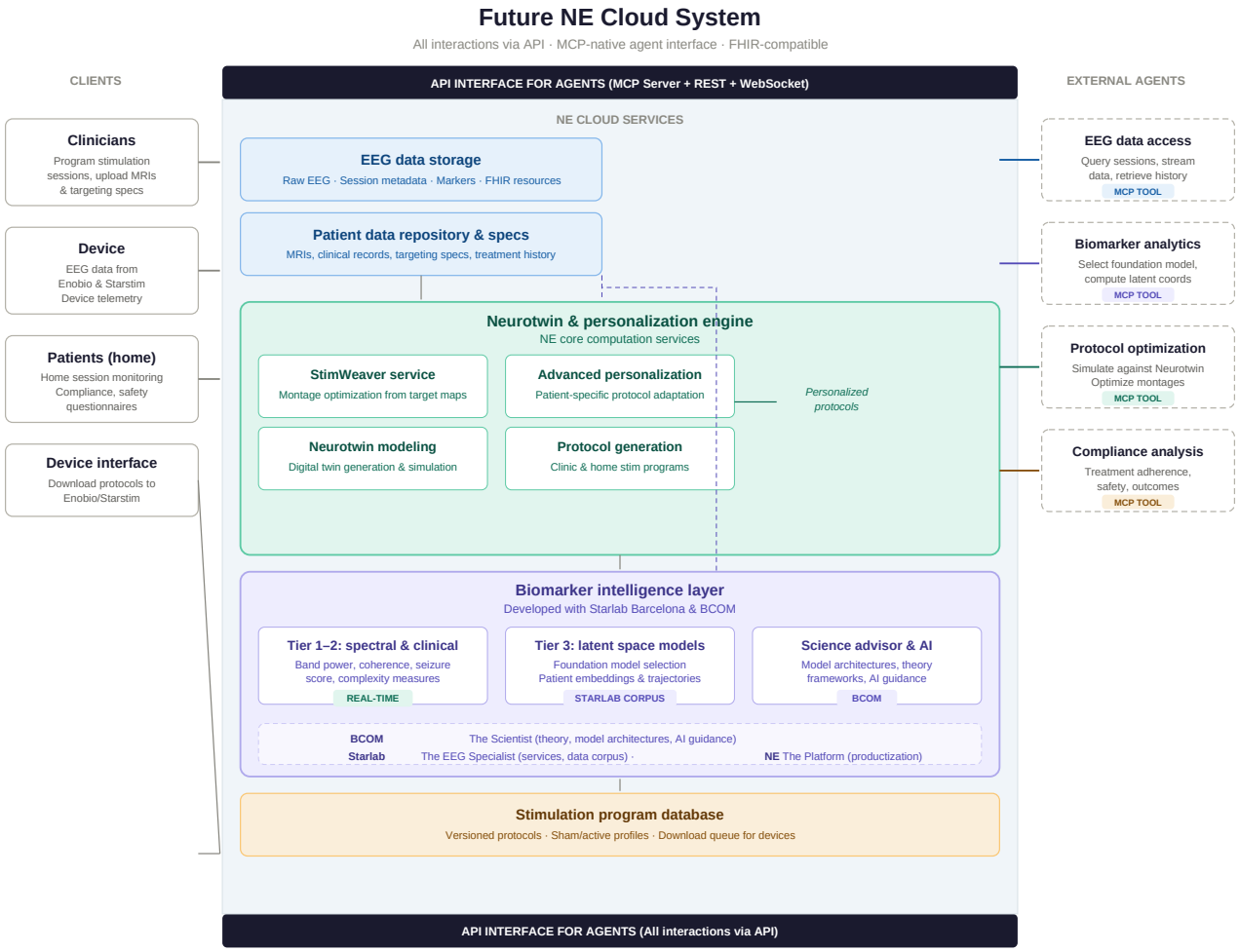


Figure 2: Future NE Cloud System architecture. Dashed borders indicate external agent access points via MCP tools. The Biomarker Intelligence Layer is developed in partnership with Starlab Barcelona (EEG corpus & foundation models) and BCOM (scientific guidance, model architectures & AI support).

This memo is intended to open a structured conversation within the Product team. A follow-up working session is recommended to prioritize the roadmap, assign ownership of the Q2 actions, and define resource requirements for Horizon 1 execution.

— End of Memo —